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United States Patent	12396562
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Ge; Wenguo

Travel pillow

Abstract

A travel pillow includes a mounting assembly and a main body. The main body includes a U-shaped case detachably connected to the mounting assembly. The mounting assembly includes a cover, a U-shaped filler, and an inner frame. The cover, the U-shaped filler, and the inner frame are disposed sequentially from outside to inside; and the U-shaped filler is detachably attached to the U-shaped case. The inner frame includes an arc-shaped support body and two arms; two ends of the arc-shaped support body include two first knuckles, respectively; and the travel pillow further includes two rotary connectors connected to the two first knuckles, respectively; each arm includes a second knuckle. The first knuckle includes a through hole; each rotary connector includes a rotating shaft and a nut. The rotating shaft includes a threaded end and a rotating part.

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Appl. No.:	18/198860
Filed:	May 18, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230284782 A1	Sep. 14, 2023

Foreign Application Priority Data

CN	202110162694.9	Feb. 05, 2021
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Related U.S. Application Data

Publication Classification

Int. Cl.: A47C7/38 (20060101); A47G9/10 (20060101)

U.S. Cl.:

CPC A47C7/383 (20130101); A47G9/1009 (20130101); A47G9/1081 (20130101);

Field of Classification Search

CPC: A47C (7/36); A47C (7/38); A47C (7/383); A47C (7/386); A47G (9/10); A47G (9/1009);
A47G (9/1081); A47G (2009/1018)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3283344	12/1965	Blanchard	5/639	A47G 9/1009
6305749	12/2000	O'Connor	5/636	B60N 2/868
6640367	12/2002	Hsia	5/636	A47G 9/1009
8898840	12/2013	Majette	5/643	A47C 7/383
9615682	12/2016	Maddocks	N/A	B60N 2/5664
2002/0185573	12/2001	Bisch	248/118	B60N 2/882
2003/0135927	12/2002	Hsia	5/643	A47G 9/1009
2004/0217639	12/2003	Clough	297/391	A47C 7/38
2017/0071349	12/2016	Wong	N/A	A47C 7/383
2020/0060450	12/2019	Wong	N/A	A47C 27/20
2020/0221887	12/2019	Jamil	N/A	A47G 9/1009

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of International Patent Application No. PCT/CN2021/135249 with an international filing date of Dec. 3, 2021, designating the United States, now pending, and further claims foreign priority benefits to Chinese Patent Application No. 202110162694.9 filed Feb. 5, 2021. The contents of all of the aforementioned applications, including any intervening amendments thereto, are incorporated herein by reference. Inquiries from the public to applicants or assignees concerning this document or the related applications should be directed to: Matthias Scholl P.C., Attn.: Dr. Matthias Scholl Esq., 245 First Street, 18th Floor, Cambridge, MA 02142.

BACKGROUND

(1) The disclosure relates to the field of household goods, and more particularly, to a travel pillow.
(2) Conventionally, a U-shaped travel pillow includes a cloth cover and an inner lining formed integrally with the cloth cover. When the cloth cover is dirty, it is difficult to clean, disassemble, and replace.

SUMMARY

(3) To achieve the aforesaid objective, the disclosure provides a travel pillow in the shape of the letter U; the travel pillow comprises a mounting assembly and a main body; and the main body comprises a U-shaped case detachably connected to the mounting assembly. The mounting assembly comprises a cover, a U-shaped filler, and an inner frame; the cover, the U-shaped filler, and the inner frame are disposed sequentially from outside to inside; and the U-shaped filler is detachably attached to the U-shaped case; the inner frame comprises an arc-shaped support body and two arms; two ends of the arc-shaped support body comprise two first knuckles, respectively; and the travel pillow further comprises two rotary connectors connected to the two first knuckles, respectively; each arm comprises a second knuckle; the first knuckle comprises a through hole; each rotary connector comprises a rotating shaft and a nut; the rotating shaft comprises a threaded end and a rotating part; the threaded end is in threaded connection to the nut; and the rotating shaft is disposed in the through hole; the first knuckle is disposed between the nut and the second knuckle; and the rotating part is disposed in the second knuckle; and the two arms are rotatably disposed on both ends of the arc-shaped support body, respectively; and an included angle between the arc-shaped support body and each arm is adjustable.

(4) The term “detachably connected” refers to a connection mean that provides detachable connectors for joining different parts together and hence allow low-cost replacement of specific parts, thus allowing for easy assembly and disassembly of the travel pillow, and allowing easy cleaning and replacement of the mounting assembly and the U-shaped case. The cover is configured to protect the U-shaped filler and the inner frame from being damaged, and cover a plurality of connectors disposed on the U-shaped filler to give an improved visual aesthetic. The U-shaped filler is configured to specify locations of the U-shaped inner frame and the cover, allowing easy assembly of the travel pillow and making the travel pillow softer. The inner frame is configured to support the main body and the U-shaped filler, thus allowing for easy assembly of the travel pillow.

(5) The arc-shaped support body is configured to prop up the user's neck; the two arms are rotated to fit a diverse range of neck shapes and transmit pressure from the arc-shaped support body to the main body, thus improving user's comfort and protecting the inner frame from damage caused by an applied force. The included angle is rotatable to protect the two arms and the arc-shaped support body from damage caused by an applied force.

(6) In a class of this embodiment, the mounting assembly is U-shaped and comprises an inner surface in contact with the U-shaped case; and the U-shaped case comprises an outer surface fitting in with the inner surface.

(7) The mounting assembly is matched with the U-shaped case in shape and detachably attached thereto.

(8) In a class of this embodiment, the cover comprises an arc-shaped middle and two end parts; and the two end parts are formed integrally with or detachably connected to the arc-shaped middle.

(9) An integral formation process reduces number of components and greatly reduces manufacturing time; and a detachable connection allows for easy maintenance of the inner frame and rapid replacement of the two arms or the arc-shaped support body.

(10) In a class of this embodiment, the U-shaped filler is detachably connected to the U-shaped case through a zipper; and the U-shaped filler comprises silicone or rubber.

(11) The silicone or rubber is stiffer than a sponge or a cloth, and enables a stable bolt connection

among the U-shaped filler, the cover, and the inner frame; the silicone or rubber improves comfort and prop for a user's neck; and the U-shaped filler is detachably connected to the U-shaped case, which allows easy cleaning or replacement of the U-shaped case.

(12) In a class of this embodiment, the U-shaped filler comprises a first groove; the arc-shaped support body comprises a first protrusion; and the first groove is detachably connected to the first protrusion.

(13) When the first groove is detachably connected to the first protrusion, the U-shaped filler is fixed on the arc-shaped support body, which allows for easy assembly and disassembly of the travel pillow.

(14) In a class of this embodiment, the U-shaped filler further comprises a second groove; each of the two arms comprises a second protrusion; the second protrusion is detachably disposed in the second groove.

(15) When the second protrusion is detachably disposed in the second groove, the corresponding arm is fixed on the U-shaped filler, which allows for easy assembly and disassembly of the travel pillow.

(16) Each rotary connector is configured to control an included angle between the arc-shaped support body and a corresponding arm; and the included angle is controlled to fit a diverse range of neck shapes, thus improving user's comfort.

(17) The following advantages are associated with the travel pillow of the disclosure; 1. The U-shaped case is detachably connected to the mounting assembly, which allows easy cleaning, disassembly, and replacement of the travel pillow; 2. The travel pillow is easy to store when not in use; 3. The two arms are rotatable, thus fitting a diverse range of neck shapes; 4. The U-shaped filler comprises silicone whose elasticity and softness enables a stable detachable connection between the inner frame and the cover, thus improving user's comfort; and 5. The travel pillow is an ergonomic pillow designed to keep user's head and spine in a neutral position, help reducing neck and shoulder pain.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a travel pillow according to one example of the disclosure;

(2) FIG. 2 is an exploded view of a travel pillow according to one example of the disclosure;

(3) FIG. 3 is a perspective view of a mounting assembly according to one example of the disclosure;

(4) FIG. 4 is an exploded view of a mounting assembly according to one example of the disclosure;

(5) FIG. 5 is a perspective view of a combination of a U-shaped filler and a U-shaped case according to one example of the disclosure;

(6) FIG. 6 is an exploded view of a combination of a U-shaped filler and a U-shaped case according to one example of the disclosure;

(7) FIG. 7 is a perspective view of a U-shaped filler according to one example of the disclosure;

(8) FIG. 8 is a perspective view of an inner frame according to one example of the disclosure;

(9) FIG. 9 is a perspective view of an arc-shaped support body in FIG. 8;

(10) FIG. 10 is a front view of an arc-shaped support body in FIG. 9;

(11) FIG. 11 is a perspective view of an arm in FIG. 8;

(12) FIG. 12 is a perspective view of a combination of an arc-shaped support body and two arms according to one example of the disclosure;

(13) FIG. 13 is an enlarged perspective view of part I in FIG. 12;

(14) FIG. 14 is an enlarged exploded view of part I in FIG. 12;

(15) FIG. 15 is a perspective view of a rotating shaft according to one example of the disclosure;

and

(16) FIG. **16** is a perspective view of another travel pillow according to one example of the disclosure.

(17) In the drawings, the following reference numbers are used: **100**. U-shaped case; **200**. Mounting assembly; **210**. Cover; **210a**. Arc-shaped middle; **210b**. End part; **220**. U-shaped filler; **220a**. Hang tab; **221**. First groove; **222**. Second groove; **230**. Inner frame; **231**. Arc-shaped support body; **231a**. First protrusion; **231b**. First knuckle; **232**. Arm; **232a**. Second protrusion; **232b**. Second knuckle; **233**. Rotating shaft; **233a**. Threaded end; **233b**. Rotating part; **234**. Nut; and **300**. Travel pillow.

DETAILED DESCRIPTION

(18) To further illustrate the disclosure, embodiments detailing a travel pillow are described below. It should be noted that the following embodiments are intended to describe and not to limit the disclosure.

(19) FIG. **1** is a perspective view of a travel pillow **300** according to Example 1 of the disclosure. The travel pillow **300** is U-shaped and comprises a notch for receiving the neck and head of a user, thus providing a natural resting position for improving neck and upper back pain of the user.

(20) As shown in FIG. **2**, the travel pillow **300** comprises a mounting assembly **200** and a main body (not shown). The main body comprises a U-shaped case **100** detachably connected to the mounting assembly **200** to allow for easy assembly and disassembly of the travel pillow **300**, thus allowing easy cleaning and replacement of the mounting assembly **200** and the main body. The term “detachably connected” refers to connection means that provide detachable connectors for joining different parts together and hence allow low-cost replacement of specific parts.

(21) Referring to FIGS. **1-2**, the mounting assembly **200** is U-shaped and comprises an inner surface in contact with the U-shaped case. The U-shaped case comprises an outer surface fitting in with the inner surface. The mounting assembly **200** is matched with the U-shaped case **100** in shape and it can be detachably attached thereto by any suitable connection means such as zippers, buttons, and clamps. The detachable connection means allow for easy assembly and disassembly of the travel pillow, allow cleaning and replacement of the main body or the U-shaped case **100**.

(22) As shown in FIGS. **3-4**, the mounting assembly **200** comprises a cover **210**, a U-shaped filler **220**, and an inner frame **230**. The cover **210**, the U-shaped filler **220**, and the inner frame **230** are disposed sequentially from outside to inside so as to protect the neck of the user. The U-shaped filler **220** is detachably attached to the U-shaped case **100**. The cover **200** is configured to protect the U-shaped filler **220** and the inner frame **230** from being damaged. The cover **210** comprises an arc-shaped middle **210a** and two end parts **210b**. The cover **210** covers a plurality of connectors (not shown) disposed on the U-shaped filler **220** to give an improved visual aesthetic. The arc-shaped middle **210a** is configured to protect the U-shaped filler **220** from being damaged. Each end part is configured to protect the corresponding arm from being damaged. The cover **210**, the U-shaped filler **220**, and the inner frame **230** are successively connected by the plurality of connectors, which allows for easy attachment and detachment of the cover **210** and the inner frame **230**. The U-shaped filler **220** is configured to improve user's comfort. The inner frame **230** is configured to support the main body and the U-shaped filler **220**, thus allowing for easy assembly of the travel pillow **300**.

(23) Referring to FIG. **8**, the inner frame **230** comprises an arc-shaped support body **231** and two arms **232**; the two arms **232** are rotatably disposed on both ends of the arc-shaped support body **231**, respectively. An included angle between the arc-shaped support body **231** and each arm is adjusted to satisfy the requirement of the user. The arc-shaped support body **231** is configured to prop up the user's neck. The two arms **232** are rotated to transmit pressure from the arc-shaped support body **231** to the main body so as to adjust an opening of the main body for receiving the neck of the user, thus improving user's comfort.

(24) Referring to FIG. **4**, the cover **210** comprises an arc-shaped middle **210a** and two end parts

210b; the two end parts **210b** are formed integrally with the arc-shaped middle **210a** or detachably connected thereto by a bolted joint or a snap-fit joint; a detachable connection allows for easy maintenance of the inner frame **230** and rapid replacement of the two arms **232** or the arc-shaped support body **231**; an integral formation is performed through a stamping process, reducing number of components and greatly reducing manufacturing time. The cover **210** comprises a rigid-plastic material or a stiff metal material for protecting the U-shaped filler **220** and the arc-shaped support body **231** from being damaged.

(25) Referring to FIGS. 5 and 6, the U-shaped filler **220** is detachably connected to the U-shaped case **100** through a zipper or a button; the zipper comprises nylon, resin, or metal; the zipper or the button is low cost, easy to manufacture, and allows for disassembly and replacement of the U-shaped filler **220** and the U-shaped case **100**; the U-shaped filler **220** comprises an elastic material such as silicone or rubber, which is stiffer than a sponge or a cloth; the elastic material enables a stable bolt connection among the U-shaped filler **220**, the cover **210**, and the inner frame **230**; as the two arms **232** rotates relative to the arc-shaped support body **231**, the elastic material stretches and does not tear easily, compared to the cloth material; the silicone or rubber improves comfort and prop for a user's neck; and the U-shaped filler **220** is detachably connected to the U-shaped case **100**, which allows easy cleaning or replacement of the U-shaped case **100**.

(26) As shown in FIG. 7, the U-shaped filler **220** comprises a first groove **221**; as shown in FIGS. 9 and 10, the arc-shaped support body **231** comprises a first protrusion **231a**; the first groove **221** is detachably connected to the first protrusion **231a** to fix the U-shaped filler **220** on the arc-shaped support body **231**, which allows for easy assembly and disassembly of the travel pillow **300**.

(27) As shown in FIG. 7, the U-shaped filler **220** further comprises a second groove **222**; as shown in FIG. 11, each of the two arms **232** comprises a second protrusion **232a**; the second groove **222** is detachably connected to the second protrusion **232a** to fix the corresponding arm **232** on the U-shaped filler, which allows for easy assembly and disassembly of the travel pillow **300**.

(28) Referring to FIGS. 9, 12, and 13, the travel pillow **300** further comprises two rotary connectors each for controlling an included angle between the arc-shaped support body **231** and a corresponding arm **232**; each end of the arc-shaped support body **231** comprises a first knuckle **231b**; and the included angle is controlled to fit a diverse range of neck shapes, thus improving user's comfort.

(29) As shown in FIGS. 12 and 13, each arm **232** comprises a second knuckle **232b**; as shown in FIGS. 14 and 15, the first knuckle **231b** comprises a through hole, and each rotary connector comprises a rotating shaft **233** and a nut **234**; the rotating shaft **233** comprises a threaded end **233a** and a rotating part **233b**; the threaded end **233a** is in threaded connection to the nut **234**; the rotating shaft **233** is disposed in the through hole; the first knuckle **231b** is disposed between the nut **234** and the second knuckle **232b**; and the rotating part **233b** is disposed in the second knuckle **232b**. As one of the two arms **232** rotates, the rotating shaft **233** rotates as well, causing the second knuckle **232b** to rotate around the rotating part **233b**; as a result, the included angle between the arc-shaped support body **231** and the corresponding arm **232** is controlled to improve user's comfort. The threaded end **233a** is in threaded connection to the nut **234** to prevent the rotating shaft **233** from sliding out of the first knuckle **231b**.

Example 2

(30) A second example of the travel pillow **300** is illustrated in FIG. 16. It is similar to Example 1, except for the following specific differences: the travel pillow **300** further comprises a hang tab **220a** comprising a hole (not shown); the hang tab **220a** is disposed on one side of the U-shaped filler **220** that props up the user's head. In use, gravity pulls down the hang tab **220a**, so that the hang tab **220** covers a zipper in the middle of the U-shaped filler **220** and the U-shaped case **100**; when not in use, the travel pillow **300** is hung on a hook through the hang tab **220a**.

(31) It will be obvious to those skilled in the art that changes and modifications may be made, and therefore, the aim in the appended claims is to cover all such changes and modifications.

Claims

1. A travel pillow, comprising: a mounting assembly; and a main body comprising a U-shaped case detachably connected to the mounting assembly; wherein: the mounting assembly comprises a cover, a U-shaped filler, and an inner frame; the cover, the U-shaped filler, and the inner frame are disposed sequentially from outside to inside; and the U-shaped filler is detachably attached to the U-shaped case; the inner frame comprises an arc-shaped support body and two arms; two ends of the arc-shaped support body comprise two first knuckles, respectively; and the travel pillow further comprises two rotary connectors connected to the two first knuckles, respectively; each arm comprises a second knuckle; the first knuckle comprises a through hole; each rotary connector comprises a rotating shaft and a nut; the rotating shaft comprises a threaded end and a rotating part; the threaded end is in threaded connection to the nut; and the rotating shaft is disposed in the through hole; the first knuckle is disposed between the nut and the second knuckle; and the rotating part is disposed in the second knuckle; and the two arms are rotatably disposed on both ends of the arc-shaped support body, respectively; and an included angle between the arc-shaped support body and each arm is adjustable.
 2. The travel pillow of claim 1, wherein the mounting assembly is U-shaped and comprises an inner surface in contact with the U-shaped case; and the U-shaped case comprises an outer surface fitting in with the inner surface.
 3. The travel pillow of claim 1, wherein the cover comprises an arc-shaped middle and two end parts; and the two end parts are formed integrally with or detachably connected to the arc-shaped middle.
 4. The travel pillow of claim 1, wherein the U-shaped filler is detachably connected to the U-shaped case through a zipper; and the U-shaped filler comprises silicone or rubber.
 5. The travel pillow of claim 1, wherein the U-shaped filler comprises a first groove; the arc-shaped support body comprises a first protrusion; and the first groove is detachably connected to the first protrusion.
 6. The travel pillow of claim 5, wherein the U-shaped filler further comprises a second groove; each of the two arms comprises a second protrusion; and the second protrusion is detachably disposed in the second groove.
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